Sensorless Brushless DC Motor Drive Based on the Zero-Crossing Detection of Back Electromotive Force (EMF) From the Line Voltage Difference

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***Abstract*—This paper describes a position sensorless operation of permanent magnet brushless direct current (BLDC) motor. The position sensorless BLDC drive proposed, in this paper, is based on detection of back electromotive force (back EMF) zero crossing from the terminal voltages. The proposed method relies on a difference of line voltages measured at the terminals of the motor. It is shown, in the paper, that this difference of line voltages provides an amplified version of an appropriate back EMF at its zero crossings. The commutation signals are obtained without the motor neutral voltage. The effectiveness of the proposed method is demonstrated through simulation and experimental results.**

***Index Terms*—Back electromotive force (EMF) detection, brushless dc (BLDC) motor, sensorless control, zero crossing.**

I. Introduction

PERMANENT MAGNET (PM) motors have been widely used in a variety of applications in industrial automation and consumer appliances because of their higher efficiency and power density. Brushless dc (BLDC) motors, with their trapezoidal electromotive force (EMF) profile, requires six discrete rotor position information for the inverter operation. These are typically generated by Hall-effect switch sensors placed within the motor. However, it is a well-known fact that these sensors have a number of drawbacks. They increase the cost of the motor and need special mechanical arrangements to be mounted. Further, Hall sensors are temperature sensitive, and hence limit the operation of the motor. They could reduce system reliability because of the extra components and wiring. Furthermore, sensorless control is the only reliable way to operate the motor for applications in harsh environments.

The BLDC motor without position and speed sensors has attracted wide attention and many papers have reported work on this. These methods are based on, using back EMF of the motor [1]–[3], detection of the conducting state of freewheeling diode in the unexcited phase [4], back EMF integration method [5], [6], detection of stator third harmonic voltage components [7], [8].

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Back EMF estimation methods typically rely on the zero-crossing detection of the EMF waveform. The technique of estimating back EMF by sensing the terminal voltages with respect to a virtual neutral point is proposed in [1]. The neutral point will not be stable during pulse width modulation (PWM) switching. Low pass filters have been used to eliminate the higher harmonics and to convert the terminal voltages into triangular waveform signals. Delay is introduced in the sensed signal due to heavy filtering, which also varies with the operating speed. Therefore, this method is well suited only for a narrow speed range. Indirect back EMF sensing technique is proposed in [2] without the need of neutral or virtual neutral potential. The back EMF zero crossing is sensed with respect to the negative dc bus potential. In [3], Kim and Ehsani define a function depending on the measured voltages, currents, and the derivative of the currents, which indicates the switching instants. After prepositioning, Kim and Ehsani [3] advance the switching pattern by 60 electrical degrees and let their sensorless algorithm take over. Since their functions are dependent on the computation of derivatives of currents, the method requires digital implementation and could be affected by sensor noise.

Detecting the free-wheeling diode conduction in the open phase gives the zero-crossing point of the back EMF waveform [4]. This approach of rotor-position sensing works over a wide speed range, especially at lower speed. The main drawback of this scheme is the requirement of six additional power supplies for the comparator circuits to detect current flowing through the free-wheeling diode.

Integrating the back EMF waveform of the unexcited phase is another method to extract the position information for the phase commutation [5]. Integration starts when zero crossing of the back EMF occurs and the integration stops when the threshold set value is reached, which gives the commutation instant. This approach is less sensitive to switching noise but low speed operation is poor. Further, this scheme needs the neutral potential and suffers from the offset error due to integration. Based on this technique, a low-cost sensorless scheme has been proposed [6]. Only one terminal voltage is sensed to detect the switching instant of a phase. Due to the interpolation of the switching instants for other two phases from the sensed phase switching instants, frequent, rapid acceleration, or deceleration is not possible.

Switching instants of star-connected BLDC motors have also been estimated from the third harmonic of the back EMF waveform [7], [8]. Summation of the terminal voltages gives the third harmonic voltage. The third harmonic voltage component

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is then integrated to find the third harmonic flux linkage whose zero crossing corresponds to the commutation instants [7]. The approach based on the inherent third harmonic voltage components [7] has the limitation that the amplitude and phase of harmonic components vary with magnetic saturation, and it is not suited to the low speed range owing to the relatively low amplitude of harmonic component voltages. Shen and Iwasaki [8] propose the integration of third harmonic of back EMF instead of terminal voltages using application specified integrated circuit (ASIC) for ultrahigh speed operation, however, access to motor winding neutral potential is required. A detailed review of the recent literature on sensorless methods is given by Acarnley and Watson [9]. Direct commutation instant detection from line voltages is proposed by Chen and Liaw [10] along with a fine-tuning technique to further improve the accuracy of the detected commutation instant. In [11], Chen and Cheng implement sensorless operation by using the average line to line voltage, which is obtained by filtering the PWM waveforms. Filtering introduces a delay, which to be minimized, requires a high switching frequency. Further, Chen and Cheng [11] do not discuss the details of sensorless starting of the motor. Cheng and Tzou in [12] designed and implemented an integrated circuit for the sensorless operation of BLDC motor by sensing the motor terminal voltages. Frequency independent phase shifter is proposed by Jung and Ha [13] for sensorless control of BLDC motor, which can shift the zero-crossing point of input signal with a specified phase delay. However, direct commutation instant detection technique proposed in [10] and [11] lacks this flexibility to advance the commutation instant, which is possible to implement using the back EMF zero-crossing detection techniques.

An extended Kalman filter estimator for a brushless dc motor has been developed by Jung and Ha [14] for speed and rotor position estimation. An obstacle to applying the extended Kalman filter algorithm to rotor position estimation is the need to set appropriate values for the covariance matrix parameters, which reflect the uncertainties in modeling and measurements. The parameter values are often chosen by trial and error. Advancing the commutation instant increases the torque production particularly at high speed operation of BLDC motor as proposed and analyzed by authors in [15] and [16]. However, direct commutation instant detection technique proposed in [10] and [11] lacks this flexibility to advance the commutation instant, which is possible to implement using the back EMF zero-crossing detection techniques.

This paper proposes a simple and reliable method for the detection of back EMF zero-crossings for sensorless operation. In this paper, the zero crossings of the back EMF are estimated indirectly from the terminal voltages measured with respect to dc negative terminal. The method does not involve any integrations. Further, since line voltages are used, the requirement of neutral potential has been eliminated. This also eliminates the common mode noise. Device drops and their variations would also not play a part since line voltages are used. Unlike the method described in [3], this scheme is easy to implement. No derivative operations are involved. In [18], the approach to zero-crossing detection was used to reliably start the BLDC machine in sensorless operation. In this paper, the approach is extended to propose a simple running mode algorithm.

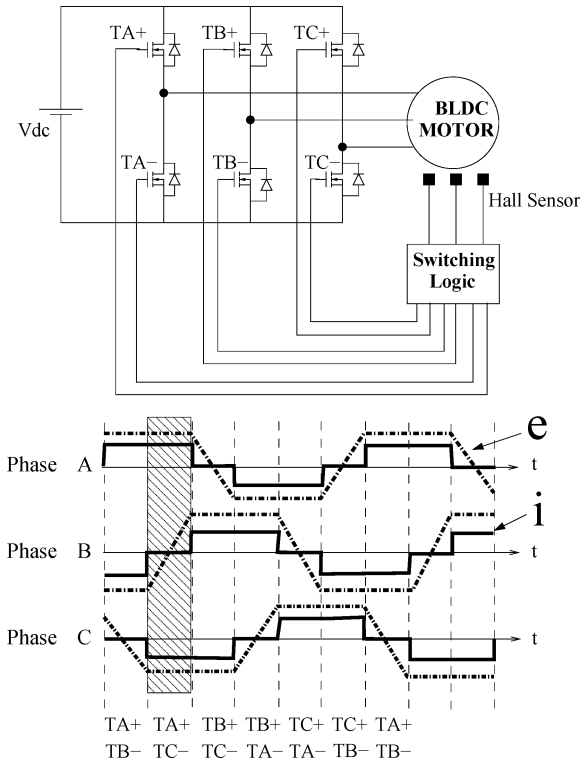


Fig. 1. BLDC motor drive along with typical phase current and back EMF.

The organization of this paper is as follows. Section II describes the proposed back EMF zero-crossing estimation method. Section III presents the simulation results of the proposed method. Section IV presents the hardware implementation results that validate the proposal and Section V presents the conclusion.

II. Proposed Back EMF Zero-Crossing Estimation Method

Consider a BLDC motor having three stator phase windings connected in star. Permanent magnets are mounted on the rotor. The BLDC motor is driven by a three phase inverter in which the devices are triggered with respect to the rotor position as shown in Fig. 1. The phase A terminal voltage with respect to the star point of the stator V_{an}, is given in (1)

$$V_{an} = R_a i_a + L_a \frac{di_a}{dt} + e_{an} \tag{1}$$

where R_a is the stator resistance, L_a is the phase inductance, e_{an} is the back EMF, and i_a is the phase current of the "A" phase.

Similar equations can be written for the other two phases, as in (2) and (3)

$$V_{bn} = R_b i_b + L_b \frac{di_b}{dt} + e_{bn} \tag{2}$$

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$$V_{cn} = R_c i_c + L_c \frac{di_c}{dt} + e_{cn} \tag{3}$$

where the symbols have their obvious meanings.

From this, the line voltage V_{ab} may be determined as

$$V_{ab} = V_{an} - V_{bn} = R(i_a - i_b) + L\frac{d(i_a - i_b)}{dt} + e_{an} - e_{bn}. \tag{4}$$

Similarly

$$V_{bc} = R(i_b - i_c) + L\frac{d(i_b - i_c)}{dt} + e_{bn} - e_{cn} \tag{5}$$

$$V_{ca} = R(i_c - i_a) + L\frac{d(i_c - i_a)}{dt} + e_{cn} - e_{an}. \tag{6}$$

These line voltages can, however, be estimated without the need for star point by taking the difference of terminal voltages measured with respect to the negative dc bus.

Subtracting (5) from (4) gives

$$V_{abbc} = R(i_a - 2i_b + i_c) + L\frac{d(i_a - 2i_b + i_c)}{dt} + e_{an} - 2e_{bn} + e_{cn}. \tag{7}$$

Consider the interval when phases A and C are conducting and phase B is open as indicated by the shaded region in Fig. 1. In this interval, phase A winding is connected to the positive terminal of the dc supply, phase C to the negative terminal of the dc supply and phase B is open. Therefore, $i_a = -i_c$ and $i_b = 0$. It can be seen from Fig. 1 (shaded region) that the back EMF in phases A and C are equal and opposite. Therefore, in that interval (7) may be simplified as

$$V_{abbc} = V_{ab} - V_{bc} = e_{an} - 2e_{bn} + e_{cn} = -2e_{bn}. \tag{8}$$

The difference of line voltages waveform is, thus, an inverted representation of the back EMF waveform. The EMF values would be those in a resistance, inductance, EMF (RLE) representation of the phase (not referred to ground). It may also be noted that the subtraction operation provides a gain of two to the EMF waveform thus amplifying it. It is again evident from Fig. 1 that during this interval (shaded portion) the back EMF e_{bn} transits from one polarity to another crossing zero. Therefore, the operation $V_{ab} - V_{bc}$ (V_{abbc}) enables detection of the zero crossing of the phase B EMF. Similarly, the difference of line voltages V_{bcca} enables the detection of zero crossing of phase C back EMF when phase B and C back EMFs are equal and opposite. The difference of line voltages V_{caab} waveform gives the zero crossing of phase A back EMF where phases C and B have equal and opposite back EMFs. Therefore, the zero-crossing instants of the back EMF waveforms may be estimated indirectly from measurements of only the three terminal voltages of the motor.

While the aforementioned discussion used an ideal trapezoidal waveform, the practical induced EMF deviates from this waveshape due to slot ripples. The validity of (8) for a practical machine is verified from experimental waveforms. Real back EMF waveforms are measured and the line voltage difference V_{abbc} is evaluated from the expression $ea+ec-2eb$ using the measured back EMF waveforms in order to verify the zero-crossing instants match. Fig. 2 shows the V_{abbc} waveform along with the back EMF waveform (multiplied by gain two, $-2eb$) and the error between the two. It is evident from Fig. 2 that the V_{abbc} waveform matches well with the back EMF waveform $-2eb$ in the zero-crossing region. The error between the two, also shown in Fig. 2, is negligible at the zero-crossing instant.

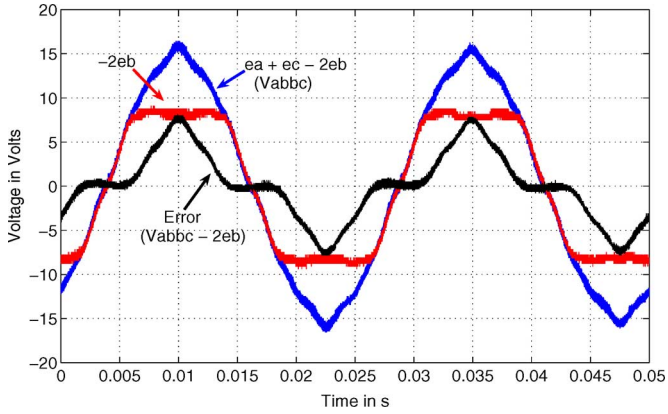


Fig. 2. Line voltage difference from experimental back EMFs.

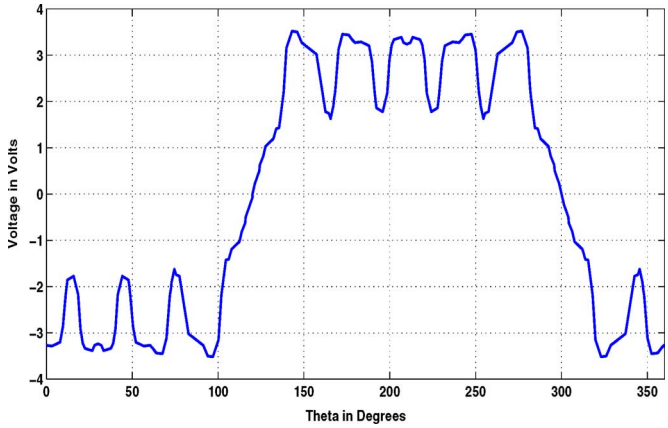


Fig. 3. Experimental back EMF waveform with higher ripple [17].

While the EMF waveform in Fig. 2 pertains to the machine used further in experimentation, the validity of (8) was further tested with another experimental EMF waveform published in the literature [17], which shows a much higher slot ripple. Fig. 3 shows this EMF waveform. Fig. 4 shows the comparison between V_{abbc} and $-2eb$ for this EMF waveform. It can be seen that the zero crossings of V_{abbc} and the back EMF of phase B match, even though slot ripples are high. The method is thus quite general and is not dependent on the ideal nature of the back EMF waveform.

The proposed sensorless method uses this approach to estimate the zero-crossing instants of the back EMF from the terminal voltages of the motor from which the correct commutation instants are estimated.

III. Simulation of the Proposed Sensorless Method

This section presents the aspects of continuous sensorless operation and the method to changeover from starting mode. Both

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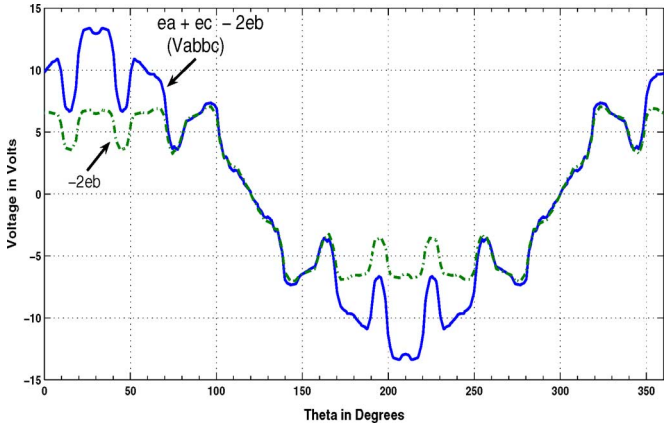


Fig. 4. Line voltage difference for the EMF waveform of Fig. 3.

TABLE I
BLDC MOTOR PARAMETERS

Parameters	Values
Voltage	12V
Resistance	0.5 Ω
Inductance	0.375m H
Back-EMF constant	0.0484 $V/rad/s$
Torque constant	0.0484 Nm/A
No. of Poles	6
Rotor Inertia	3.989m kgm^2

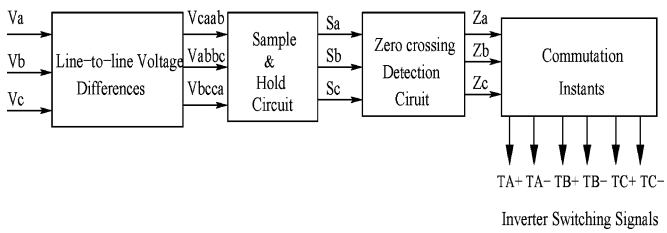


Fig. 5. Functional sequence of operations in proposed algorithm.

starting and subsequent operations make use of the ideas presented in the previous section. Sensorless BLDC motor starting technique proposed in [18] is used to start the motor.

The proposed sensorless method is simulated in SABER software. The motor parameters used for simulation and hardware implementation are given in Table I. The functional sequence of operations of the proposed back EMF zero-crossing detection for sensorless operation is shown in Fig. 5, and is further explained in the following.

From the sensed terminal voltages with respect to negative dc bus (V_a, V_b, V_c), line voltages, and subsequently their differences (V_{caab}, V_{abbc}, V_{bcca}) are determined. Fig. 6 shows the simulated back EMF waveform of phase B and the line voltage difference V_{abbc}. Fig. 6 further shows that the zero-crossing instants of V_{abbc} match with that of the phase B back EMF e_b. These are indicated as zero-crossing points (ZCP) in Fig. 6. It can be seen that the plot validates (8) in the region of zero crossing of the back EMF. Therefore, we conclude that the method of detection of zero crossings of the back EMF, as described in the previous section, is valid.

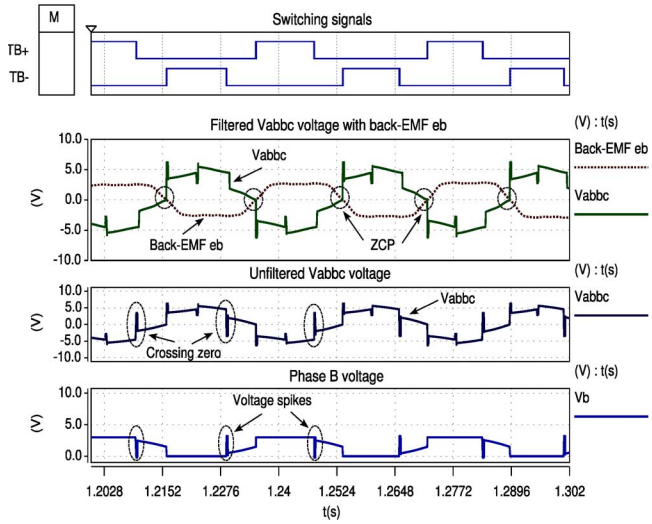


Fig. 6. Detection of back EMF zero crossing.

Fig. 6 also shows the terminal voltage V_b of phase B. This waveform contains spikes due to free-wheeling diode conduction. When the line voltage difference, say V_{abbc}, is found, these spikes cause false zero-crossing detection, and hence need to be eliminated. This is done by holding the value of the terminal voltage before spike occurrence for a predetermined duration. A detailed analysis justifying this technique is given in the Appendix. This operation is shown as the sample and hold block in Fig. 5.

Once the motor is started in the sensorless starting mode [14], then the correct commutation instants are to be estimated from the ZCP in order to switch over to running mode. The flow chart to estimate the correct commutation instants from the proposed back EMF zero-crossing detection method is shown in Fig. 7. The sensorless commutation instants are delayed by 30 electrical degrees from the back EMF zero-crossing instants in order to excite the phase windings during the flat portion of the back EMF (see Fig. 1). This is done as follows. When the motor is in the sensorless starting mode, a 12-bit digital counter is started at a zero-crossing instant of the back EMF. Zero crossings occur every 60 electrical degrees of the rotor rotation. The timer is stopped at the subsequent zero-crossing instant. In the flowchart given in Fig. 7, these are the zero-crossing instants of the back EMF in phases C and B, respectively. Thus, the duration of a 60° interval is established. Half of this time interval, which is 30°, is obtained by a right shift of the counted value by 1 bit, and then it is loaded into a digital comparator. The timer counter of the comparator is then started. The instant when the timer counter value of the comparator equals the loaded count for 30° is taken to be the commutation instant of a device. For the flowchart given in Fig. 7, this would be the instant of turn on of TB+ and turn off of TA+. Table II shows the sequence in which the counter for 60 electrical degrees functions for each of the device switching signals.

In the simulation results shown here, the motor is started from standstill and runs in starting mode till 165 ms and then changes over to sensorless mode. Fig. 8 shows the zero-crossing

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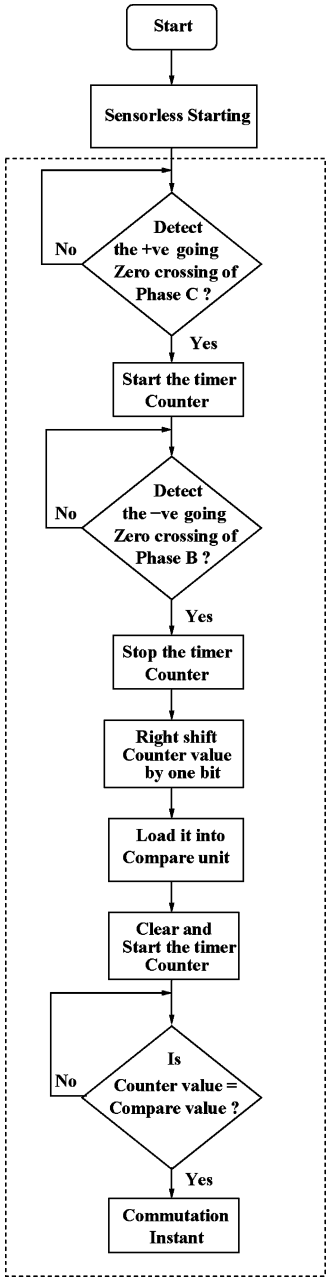


Fig. 7. Flow chart for change over from starting to running mode.

TABLE II
SEQUENCE FOR 60 DEGREES COUNTER OPERATION

Device Switching signal	Counter for 60 degree period	
	Start at	Stop at
TA+	+ve going ZCP of Vabbc	-ve going ZCP of Vcaab
TC-	-ve going ZCP of Vcaab	+ve going ZCP of Vbcca
TB+	+ve going ZCP of Vbcca	-ve going ZCP of Vabbc
TA-	-ve going ZCP of Vabbc	+ve going ZCP of Vcaab
TC+	+ve going ZCP of Vcaab	-ve going ZCP of Vbcca
TB-	-ve going ZCP of Vbcca	+ve going ZCP of Vabbc

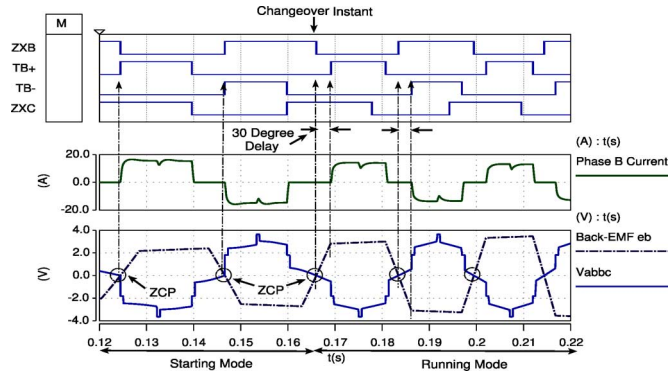


Fig. 8. Estimation of commutation instants from ZCP for phase B.

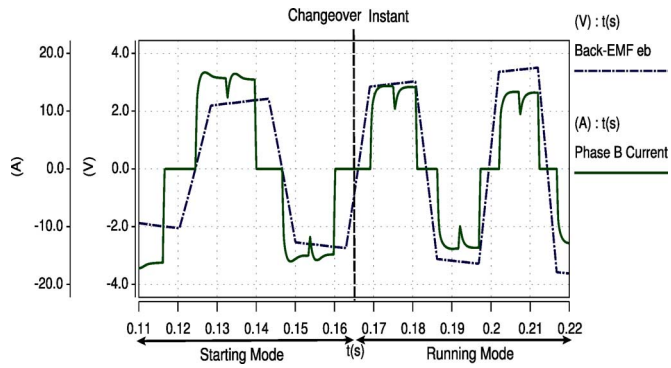


Fig. 9. Phase B back EMF and current during starting and running mode.

signal ZXB of difference of line voltages V_{abbc} just before and after changeover to running mode. The gating signals for TB+ and TB− are also shown. It can be seen that the devices are triggered on in synchronous with transitions in ZXB while in starting mode. In running mode there is a 30° delay. The back EMF along with current of phase B during starting and running modes is shown in Fig. 9. It is evident from Fig. 9 that the commutations are performed at the ZCPs in the starting mode, whereas in the running mode, it is in phase with the flat portion of the back EMF. Subsequent to the changeover, 60° interval detection and the delay of 30° is performed on a continuous basis to maintain operation in the running mode.

IV. HARDWARE IMPLEMENTATION AND TEST RESULTS

Fig. 10 shows the block diagram of the experimental setup of the proposed sensorless BLDC motor drive. The experimental setup consists of a 400 W, 3000 r/min, six pole BLDC motor drive coupled to a dc generator connected to a variable resistor for loading. The parameters of the motor are the ones used in the simulation study and listed in Table I. The proposed sensorless method is implemented using TI TMS320LF2407A digital signal processor (DSP). The BLDC motor is driven by a three-phase power MOSFET module. The three terminal voltages of the

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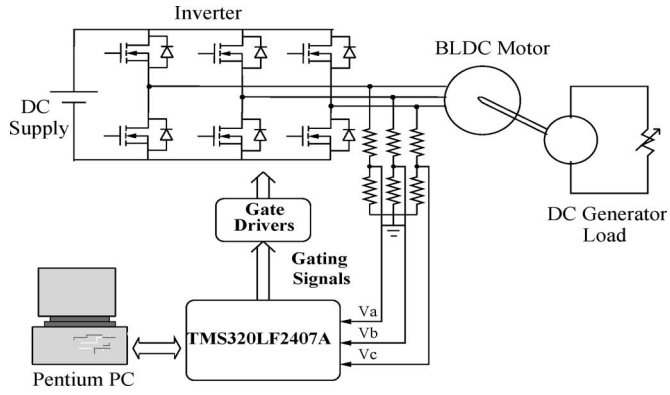


Fig. 10. Block diagram of the experimental setup.

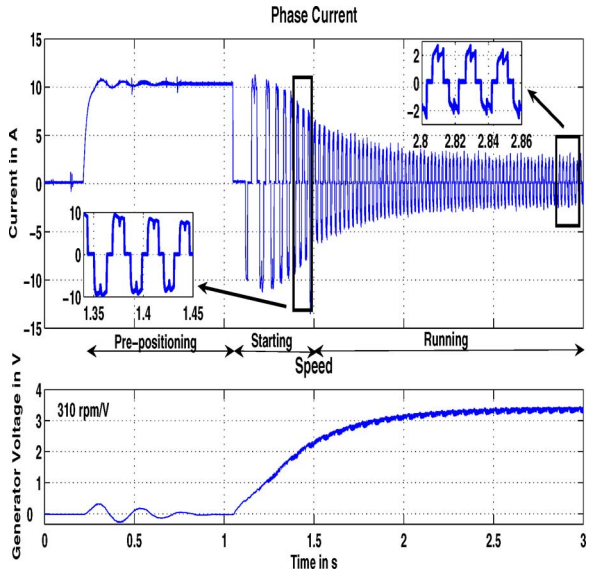


Fig. 11. Phase current and speed waveform on no-load (experimental).

motor are sensed and fed to the analog-to-digital converter (ADC) of the DSP through a resistive divider as shown in Fig. 10 in order to reduce the magnitude. The voltage spikes that appear in the sensed terminal voltage waveform due to the free-wheeling diode conduction during phase commutation are filtered out by the sample and hold method as discussed in Section III. The high-frequency switching noise due to PWM is filtered out by synchronizing the ADC sampling such that sampling is done only during the PWM on time.

Once the motor is started, in order to run the motor in the sensorless mode the switching signals need to be delayed by 30 electrical degrees from the ZCP of the back EMF. A digital delay algorithm is used in the DSP as shown in Fig. 7 and as explained in Section III. Fig. 11 shows the phase B current and the speed waveform from starting mode to running mode. The rotor prepositioning lasts for 1.1 s (T_p is 800 ms). Subsequently, starting mode lasts till 1.5 s. Changeover occurs at 1.5 s wherein the operation is changed to running mode. The zoomed view of current waveforms during the starting mode and running mode are also shown in Fig. 11. The speed rises steadily from standstill and reaches the steady state without any serious distortion as seen in Fig. 11.

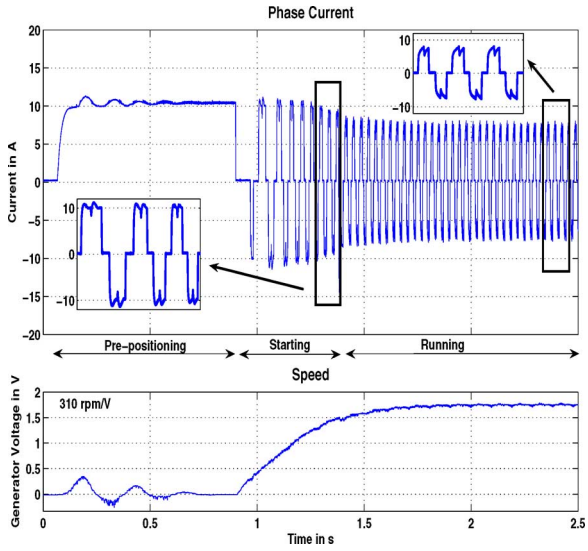


Fig. 12. Phase current and speed waveform on load (experimental).

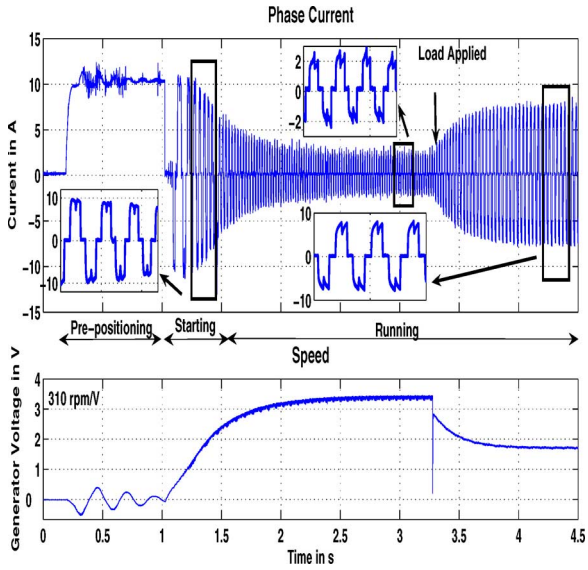


Fig. 13. Phase current and speed waveform during loading transient (experimental).

The phase current and speed of the motor on starting with loaded condition are shown in Fig. 12. Prepositioning of the rotor takes place till 0.9 s (T_p is 800 ms). The motor operates in the starting mode till 1.35 s, and then changes over to running mode. Fig. 13 on the other hand shows the current and speed waveforms with the motor started on no-load but is subsequently loaded in the running mode. A high load is suddenly applied to the motor during the sensorless mode at 3.3 s to test the dynamic performance of the proposed sensorless algorithm. That the motor continues to run sensorless even after a sudden application of high load is evident from Fig. 13, thus validating the performance of the proposed sensorless algorithm even for a transient load condition. The virtual Hall sensor signal from the proposed sensorless algorithm matches well with the real hall sensor signal during the high-frequency switching operation as is shown in Fig. 14.

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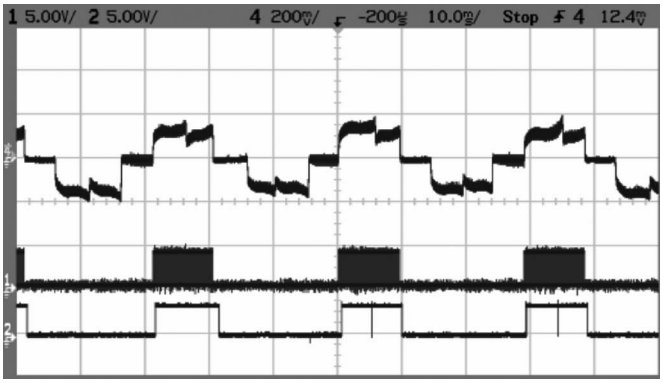


Fig. 14. Phase current, virtual Hall, and real Hall sensor signal for 50% duty ratio PWM switching.

V. Conclusion

A simple technique to detect back EMF zero crossings for a BLDC motor using the line voltages is proposed. It is shown that the method provides an amplified version of the back EMF. Only three motor terminal voltages need to be measured thus eliminating the need for motor neutral voltage. Running the machine in sensorless mode is then proposed, in this paper, making use of the novel zero-crossing detection algorithm. While starting relies on triggering devices at the zero crossings detected using the proposed algorithm, continuous running is achieved by realizing the correct commutation instants 30° delay from the zero crossings. The motor is found to start smoothly and run sensorless even with load and load transients. Simulation and experimental results are shown which validate the suitability of the proposed method.

Appendix

Consider the commutation instant when the device TA+ is turned OFF and TB+ is turned ON. Devices TA+ and TC− were conducting before this commutation instant. When device TA+ is turned OFF, diode DA− conducts to free-wheel the phase current i_a. During this free-wheeling period, the devices TB+ and TC− are ON along with the diode DA− as shown in Fig. 15. It may be noted that this duration of free-wheeling *is* the duration for which the spikes lasts, and this duration would not extend near the instant where zero crossing would take place. This can be seen in the following analysis.

Differential equations (9) and (10) describe the system as shown in Fig. 15

$$-V_{\rm dc} + R(i_b) + L\frac{d(i_b)}{dt} + e_b - e_a - L\frac{d(i_a)}{dt} - R(i_a) = 0 \tag{9}$$

$$R(i_a) + L\frac{d(i_a)}{dt} + e_a - e_c - L\frac{d(i_c)}{dt} - R(i_c) = 0. \tag{10}$$

Further

$$i_a + i_b + i_c = 0, \quad i_c = -i_a - i_b. \tag{11}$$

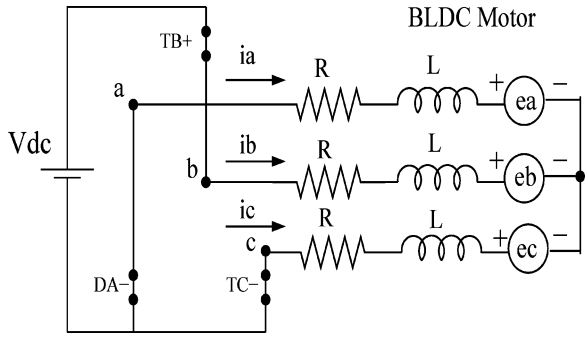


Fig. 15. Circuit diagram during free-wheeling period (DA−, TB+, and TC− are ON).

Substituting the value of i_c in (10) and rearranging it, gives (12)

$$2\left(R(i_a) + L\frac{d(i_a)}{dt}\right) + R(i_b) + L\frac{d(i_b)}{dt} + e_a - e_c = 0. \tag{12}$$

Rearranging (9) as in the following and substituting it in (12) gives (14)

$$R(i_b) + L\frac{d(i_b)}{dt} = V_{\rm dc} + e_a - e_b + R(i_a) + L\frac{d(i_a)}{dt} \tag{13}$$

$$R(i_a) + L\frac{d(i_a)}{dt} = \frac{e_b + e_c - 2e_a - V_{\rm dc}}{3}. \tag{14}$$

Assuming that back EMF during this period in phase B and phase C, respectively, are equal and opposite (14) reduces to (15)

$$R(i_a) + L\frac{d(i_a)}{dt} = \frac{-2e_a - V_{\rm dc}}{3}. \tag{15}$$

An ideal back EMF waveform has 60 electrical degree transition internal. The variation of e_a (back EMF in phase A) during this period may be written as in (16) which is further simplified to (17)

$$e_a(\theta) = E - \frac{2E}{\pi/3}\theta = E\left(1 - \frac{6\theta}{\pi}\right) \tag{16}$$

$$e_a(t) = E\left(1 - \frac{12t}{T}\right) \tag{17}$$

where $T = 2\pi/\omega$ and ω is the peed of the machine in (electrical) rad/s and E is the level of the EMF, where the waveform (ideal) is flat. E can be written as in (18)

$$E = \frac{V_{\rm dc}}{2} - I_0 R \tag{18}$$

where I_0 is the load current.

Substituting (18) in (17) gives the variation of e_a during this period

$$e_a(t) = \left(\frac{V_{\rm dc}}{2} - I_0 R\right)\left(1 - \frac{12t}{T}\right). \tag{19}$$

Substituting for e_a from (19) in (15) gives

$$R(i_a) + L\frac{d(i_a)}{dt} = -\frac{2}{3}\left(V_{\rm dc} - I_0 R\right) - \left(\frac{8I_0 R - 4V_{\rm dc}}{T}\right)t. \tag{20}$$

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Equation (20) is of the form as given in the following and the solution for current i_a is given by (23)

$$R(i_a) + L\frac{d(i_a)}{dt} = kt - \frac{2}{3}(V_{\rm dc} - I_0 R) \tag{21}$$

where k is as given in the following:

$$k = -\frac{(8I_0R - 4V_{\rm dc})}{T}. \tag{22}$$

The free-wheeling current i_a during the phase commutation is described by the following:

$$i_a(t) = \frac{kt}{R} + \left[\frac{Lk}{R^2} + \frac{2(V_{\rm dc} - I_0 R)}{3R}\right](e^{-t/\tau} - 1) + I_0 e^{-t/\tau} \tag{23}$$

where τ is the time constant equal to *L/R*.

A plot of i_a versus time, parameterized over different load currents shows that this current decays well before the zero crossing of the back EMF. As an example, for the machine discussed in the paper, the range of the free-wheeling duration is 0.5 μs to 0.55 ms. Whereas, the 30 electrical degree interval (at which time from the initiation of free-wheeling, the back EMF crosses zero) is much larger than this duration (which is about 3.36 ms). The range of currents considered for this analysis is 1–10 A at rated speed.

In order to estimate the zero-crossing instant accurately, we need to know the line voltages *around the zero-crossing region*. The aforementioned analysis shows that the free-wheeling, and hence the spike is completed well before the back EMF crosses zero. Therefore, we *can* have an upper bound for the sample and hold duration without affecting the actual values of difference of line voltages required for zero-crossing estimation. Hence, the proposed approach would work.

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